D 10673 **(Pages : 3)** **Name.....**

Reg. No.....

FIFTH SEMESTER U.G. DEGREE EXAMINATION, NOVEMBER 2021

(CBCSS–UG)

Mathematics

MTS 5D 03—LINEAR MATHEMATICAL MODELS

(2019 Admissions)

Time : Two Hours Maximum : 60 Marks

Section A

Answer at least **eight** *questions.*
Each question carries 3 *marks.*
All questions can be attended.
Overall Ceiling 24.

1. Does the line $y = -x + 5$ intersect the point $(3, -1)$? Why ?
2. Let $g(x) = -4x + k$ where k is a constant. If $g(3) = 5$, find the value of k.
3. Solve the system of equations $3x + y = 5$, $3x = 6$.
4. Write the augmented matrix for the system of equations $3x + y = 6$, $2x + 5y = 15$.
5. Find values of x, y if $\begin{bmatrix} 3 & 4 \\ -8 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 2y \\ x & 1 \end{bmatrix}$.
6. Find the product of matrices : $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \times \begin{bmatrix} -1 & 5 \\ 0 & 3 \end{bmatrix}$.
7. Graph the linear inequality $x \leq 3y$.
8. Define the term *corner point*. State the corner point theorem.
9. Give an example for a maximization problem in standard form with 2 variables.
10. Sketch the feasible region for the linear programming problem :
    Maximize $Z = 2x + 3y$ subject to $x \geq 0, y \geq 0$.
11. What are the conditions to be satisfied to call a linear programming problem to be in standard minimum form ?

Turn over

12. Write the matrix form of the linear programming problem :

Minimize $W = 8y_1 + 16y_2$

subject to $y_1 + 5y_2 \geq 9$
$2y_1 + 2y_2 \geq 10$
$y_1 \geq 0, y_2 \geq 0.$

(8 × 3 = 24 marks)

Section B

Answer at least **five** *questions.*
Each question carries 5 *marks.*
All questions can be attended.
Overall Ceiling 25.

13. In recent years, the percentage of the U.S. population age 18 and older who smoke has decreased at a roughly constant rate, from 24.1% in 1998 to 20.6% in 2008. Find the equation describing this linear relationship.

14. Solve the system of equations :

$3x + 10y = 115$
$11x + 4y = 95$

using echelon method.

15. A convenience store sells 23 sodas one summer afternoon in 12-, 16-, and 20-*oz* cups (small, medium, and large). The total volume of soda sold was 376 *oz*. Suppose that the prices for a small, medium, and large soda are \$1, \$1.25, and \$1.40, respectively, and that the total sales were \$28.45. How many of each size did the store sell ?

16. Solve the following system of equations using the inverse of the coefficient matrix :

$x + 3y - 2z = 4$
$2x + 7y + 3z = 8$
$3x + 8y + 5x = -4.$

17. Graph the feasible region for the following system of inequalities and tell if it is bounded or unbounded :

$3x - 2y \geq 6$
$x + y \leq -5$
$y \leq -6$

18. Find the maximum value of the objective function $z = 3x + 4y$, subject to the constraints : $-x + 2y \leq 4, x \geq 0, y \geq 0.$

19. Add slack variables to the following linear programming problem and write the initial simplex tableau:

Maximize $Z = 3x_1 + 2x_2 + x_3$

subject to $2x_1 + x_2 + x_3 \leq 150$
$2x_1 + 2x_2 + 8x_3 \leq 200$
$2x_1 + 3x_2 + x_3 \leq 320$
and $x_1, x_2, x_3 \geq 0$.

(5 × 5 = 25 marks)

Section C

Answer any **one** *question.*
The question carries 11 *marks.*

20. Assume that the demand of an item A increases as the price decreases. If the weekly demand for A is q (in dollar \$) and the price is p then suppose that they are related by the linear relation $D(p) = 9 - 0.75q$.

(a) Find the quantity demanded at a price of \$ 5.25 per item and at a price of \$ 3.75 per item.

(b) It is also noticed that the quantity of item A supplied decreased as the price decreased. If price p and supply q are related by the linear function $S(p) = 0.75q$, find the quantity supplied at a price of \$ 5.25 per item and at a price of \$ 3.00 per item.

(c) Graph both functions $D(p)$ and $S(p)$ on the same axes.

21. Solve the problem using simplex method :

Maximize $Z = 12x_1 + 15x_2 + 5x_3$

subject to $2x_1 + 2x_2 + x_3 \leq 8$
$x_1 + 4x_2 + 3x_3 \leq 12$
$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0.$

(1 × 11 = 11 marks)

D 10673-A **(Pages : 4)** **Name.....**

Reg. No.....

FIFTH SEMESTER U.G. DEGREE EXAMINATION, NOVEMBER 2021

(CBCSS—UG)

Mathematics

MTS 5D 03—LINEAR MATHEMATICAL MODELS

(2019 Admissions)

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes	**Total No. of Questions : 15**	**Maximum : 15 Marks**

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 15.
2. The candidate should check that the question paper supplied to him/her contains all the 15 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MTS 5D 03—LINEAR MATHEMATICAL MODELS

(Multiple Choice Questions for SDE Candidates)

1. What is the slope of line passing through the points (0,1) and (1,1) ?

   (A) 1. (B) –1.

   (C) 0. (D) 0.5.

2. Equation of a line whose x-intercept is 1 and y-intercept is –1.

   (A) $x + y = 1$. (B) $x - y = 1$.

   (C) $x + y = -1$. (D) $x - y = -1$.

3. P : Slope of a horizontal line is 1.

   Q : Slope of a vertical line is undefined.

   (A) P is true and Q is false. (B) P is false and Q is true.

   (C) P and Q are false. (D) P and Q are true.

4. Find the equation of the line that passes through the point (3, 5) and is parallel to the line $2x + 5y = 4$.

   (A) $5x + 2y = 25$. (B) $2x - 5y = -19$.

   (C) $2x + 5y = 31$. (D) $2x + 5y = -19$.

5. Find the equation of the line that passes through the point (0,1) and is perpendicular to the line $x = 0$.

   (A) $y = 0$. (B) $y = 1$.

   (C) $y = -1$. (D) None of the above.

6. What is the solution of the system of linear equation given below ?

   $x + y = 1$
   $x - y = -1$

   (A) $(0, 1)$. (B) $(0, -1)$.

   (C) $(1, 0)$. (D) $(-1, 0)$.

7. $x - 3y = 1$
$x - 3y = -1$ the linear system of equation has :

(A) Unique solution. (B) Infinitely many solutions.

(C) No solution. (D) None of the above.

8. $2x - 6y = 4$
$3x - 9y = 6$ the linear system of equation has :

(A) Unique solution. (B) Infinitely many solutions.

(C) No solution. (D) None of the above.

9. $\begin{bmatrix} 5 & -6 \\ 8 & 9 \end{bmatrix} + \begin{bmatrix} -4 & 6 \\ 8 & -3 \end{bmatrix} =$ _____.

(A) $\begin{bmatrix} -1 & 10 \\ 16 & 6 \end{bmatrix}$. (B) $\begin{bmatrix} 1 & 0 \\ 16 & 6 \end{bmatrix}$.

(C) $\begin{bmatrix} -1 & 12 \\ 16 & -6 \end{bmatrix}$. (D) $\begin{bmatrix} -1 & 0 \\ 16 & -6 \end{bmatrix}$.

10. Let A be a matrix of order $m \times n$ and B be a matrix of order $p \times q$. Then A B is defined if and only if :

(A) $m = p$. (B) $m = q$.

(C) $n = p$. (D) $n = q$.

11. $A = \begin{bmatrix} 5 & -6 \\ 8 & 9 \end{bmatrix}$ and $B = \begin{bmatrix} -4 & 6 \\ 8 & -3 \end{bmatrix}$. Then AB = _____.

(A) $\begin{bmatrix} -68 & 48 \\ 40 & 19 \end{bmatrix}$. (B) $\begin{bmatrix} -68 & 48 \\ -104 & 21 \end{bmatrix}$.

(C) $\begin{bmatrix} -68 & 42 \\ 40 & 21 \end{bmatrix}$. (D) $\begin{bmatrix} -68 & 48 \\ 40 & 21 \end{bmatrix}$.

12. If $A = \begin{bmatrix} 1 & 3 \\ 2 & 5 \end{bmatrix}$ then $A^{-1} =$ ————————.

(A) $\begin{bmatrix} -5 & 3 \\ -2 & -1 \end{bmatrix}$.

(B) $\begin{bmatrix} 5 & 3 \\ -2 & -1 \end{bmatrix}$.

(C) $\begin{bmatrix} -5 & 3 \\ 2 & -1 \end{bmatrix}$

(D) None of the above.

13. Let A be a square matrix.

P : A^{-1} exist for every square matrix A. Q : There are square matrices A whose inverse does not exist.

(A) P is true and Q is false.

(B) P is false and Q is true.

(C) P and Q are false.

(D) P and Q are true.

14. The maximum value of the objective function $z = 3x + 4y$, subject to the following constraints

$$2x + y \leq 4$$
$$-x + 2y \leq 4$$
$$x \geq 0$$
$$y \geq 0$$

(A) 6.

(B) 12.

(C) 18.

(D) 24.

15. What is the solution of the following linear programming problem ?

Minimize $Z = 2x + 4y$

subject to $x + 2y \geq 10$

$3x + y \geq 10$

$x \geq 0$

$y \geq 0$.

(A) 5.

(B) 10.

(C) 15.

(D) 20.